

1SAR artefact audit

The archive does not contain the model its report describes

- Seven rounds were run · round 7 was reported · round 4 was shipped as 1sar_final.pdb
- The round-7 coordinates are absent from the archive entirely
- The agent log attributes R-free 0.1989 (reported as 0.199) to round 7
- Five earlier review passes compared the report against the wrong model

The archive establishes a packaging mismatch; without the round-7 model or raw result it cannot adjudicate same-model reporting accuracy. The log still exposes internal defects.

What the archive actually contains

Every row below is from the artefact's own provenance

Model	R-work	R-free	In the archive?	Provenance
packaged 1sar_final.pdb	0.1563	0.2114	yes — labelled final	round-4 output
round 5	0.1555	0.2109	yes	
round 6	0.1539	0.2068	yes	
round 7 — the reported final	0.1488	0.1989	NO — coordinates absent	retained log only

Evidence

- iter4_transcript.json records the copy that made the packaged file: cp 1sar_final_001.pdb → data/1sar_final.pdb
- Its header (0.1563 / 0.2114) matches data/1sar_round5_input.pdb — the packaged “final” is what round 5 was handed
- claudе_іterations.log:510 records round 7 at R-work 0.1488, R-free 0.1989, 159 waters, 1 Ca²⁺, 1 Na⁺, 1 SO4
- claudе_іterations.log:752 names /agent/job/1sar_round7_001.pdb as the final deliverable (1654 atoms)
- 0.1989 / 0.1488 appear in four provenance transcripts; iter10 runs model_vs_data against the round-7 path
- An ignore-independent search of every archive member finds no round-7 PDB or MTZ

The retained log attributes 0.1989 to the missing round-7 path, which rounds to 0.199. Without coordinates or raw output that value is traceable, not independently verified. The archive does establish that a three-round-stale file was shipped as the final model.

The report's numbers, re-examined

The clearest surviving defect is not the one we filed

Log-traceable, not verified

The retained log attributes 0.1989 / 0.1488 to round 7. Its coordinates and raw tool result are absent, so neither value can be independently re-measured.

A proven packaging mismatch

The named round-7 file was not packaged; the shipped file is round 4. Its 0.2116 is a cross-model comparison, not a same-model accuracy test.

An internal logged defect

The logged values imply $0.1989 - 0.1488 = 0.0501$, failing the < 0.05 criterion. The report states “gap=0.050”; this does not verify either input value.

What five review passes got wrong

This harness filed, and defended across five rounds, the conclusion that 0.199 “corresponds to no model the agent produced”. That conclusion came from comparing a reported number against a file whose name said final, without once checking the file's provenance. The decisive check was object identity — not another comparison of the same headline numbers, which is what each round performed.

The archive makes the number traceable to a round-7 path, but does not independently verify it.
The defensible correction is provenance-scoped, not categorical exoneration.

What was measured — and on which model

Candidate-final rows audit packaged round 4; contextual baselines are labelled

Task	Metric	Value	Reading
T03	R-work / R-free, phenix.model_vs_data	0.1564 / 0.2116	matches packaged header
T03	R-work / R-free, gemmi matched radii	0.1622 / 0.2136	independent code path
T03	gemmi – PHENIX R-work offset	+0.0058	unmatched estimator; informational
T03	R-work/R-free gap	0.0552	no criterion applied
T14	candidate residues scored — reduce / reduce2	14 / 18	historical 0/14; calls absent
T16	DockQ vs deposited, AB:AB	0.9445	crystallographic dimer

T03 is NOT settled. driving_example_T03 is conjunctive — “all must pass for green”. Only the R-value rules were run. Rule 1's deposition reference does not exist for 1SAR (the 1990 entry records FREE R VALUE : NULL); rule 4 was not re-run; and rule 3 does not hold as written — input NREF 7248 against 7262 in every output MTZ (see #584).

The archive proves the packaging mismatch, but not the missing model's numeric accuracy. Raw packaged-model oracle output was not committed, so 0.1564 / 0.2116 need a licensed PHENIX rerun to reproduce; round 7's numbers are log-attested only.

Three findings about the artefact

The first is a defect in our own benchmark

1 · The wrong protein identity came from our own prompt

The report calls 1SAR staphylococcal nuclease; it is ribonuclease Sa. But the harness prompt says so first — `claude_iterations.log:5`: “Perform crystallographic refinement and validation of staphylococcal nuclease (PDB: 1SAR)”. Line 21 primes the Ca^{2+} site, and the supplied `1sar.pdb` carries no HEADER/TITLE metadata, so the prompt was the only identity cue.

The agent failed to challenge a false premise it was handed. It did not originate it.

2 · Both ions sit on deposited water sites

On the packaged model, HOH A164 is 0.18 Å from CA A98 and HOH B143 is 0.31 Å from NA B98. The starting model has no waters, so these peaks may be restored or retyped deposited waters rather than new ions. The ion assignments are not established either way.

3 · The starting model had two sulfates

`data/1sar.pdb` contains SO4 A97 and B97 — as ATOM records. The agent inherited chain A, kept the convention, and removed chain B, exactly as its report says. Record type is not chemical identity; this harness miscounted it before noticing.

What this changes about the May assessment

May concluded the ion decisions were defensible and the refinement sound. Neither survives: the ion sites coincide with deposited waters, the protein-specific reasoning rests on an identity our prompt supplied, and the sulfate edit the report describes was correct. Those conclusions are withdrawn pending re-measurement.

Coverage on the packaged artefact

8 tasks measured · T03 incomplete · 2 uncovered

Measured — May, carried, not re-run - T01 superposition - T05 geometry - T06 model-vs-data - T10 ligand/site - T13 data quality
Measured — this audit; candidate-final rows use packaged round 4 - T03 X-ray refinement — R-value rules only; rubric NOT satisfied - T14 hydrogen placement - T16 interface quality
Not applicable - T04 real-space refinement · T07 predicted-model processing · T08 docking into a map · T12 cryo-EM map quality · T17 NMR ensembles — none apply to an X-ray artefact - T09 molecular replacement — the model was already placed - T11 loop fitting — no chain gaps (A 1–96, B 1–96)
Uncovered - T02 per-residue comparison — driver exists; not run or retained in this audit - T15 structural classification — historical numbers lack retained inputs, content and assignments

No task is claimed as settled for the agent's reported final model — that model is not in the archive, so nothing here measures it.

Correction history

What this record withdraws, and why

Claim	Status	Why
“0.199 corresponds to no model the agent produced”	withdrawn — overstated	Log attributes 0.1989 to absent round-7 path; raw result absent
“The report describes the wrong protein” (agent's error)	reattributed	The harness prompt supplied the identity
“The starting model has zero heteroatoms”	withdrawn — false	It has two sulfates, written as ATOM
“Ion decisions remain defensible” (May)		Both ions sit on deposited water sites
“T03 settled / refinement sound”		Only the R-value rules of a conjunctive rubric were run
Five successive gradings of the R-free claim	superseded	All compared the report against the wrong model

The shared blind spot

Five review passes treated three things as ground truth that were not: the filename 1sar_final.pdb, the ATOM/HETATM record type as a proxy for chemical identity, and the harness prompt's statement of what protein this is. Each pass re-examined the same headline numbers more carefully than the last. None asked whether the object being measured was the object under discussion.

The check that resolved it — following provenance through the archive — cost minutes.

Trust and reporting caveats

What this record's own numbers can and cannot support

No quantitative oracle row carries a verdict

The gemmi R-work row uses a bin-rescaled estimator that the raw-sfcalc benchmark did not measure, and the R-free row has no test-set-specific band, so both are informational. The cctbx rows measure packaged round 4 only; round-7 values remain narrative log provenance, not structured row comparisons.

The round-7 numeric claim is not oracled

Raw packaged-model oracle output was not committed, so 0.1564 / 0.2116 need a licensed PHENIX rerun to reproduce. Round 7's 0.1488 / 0.1989 are attested only by the agent log; the archive directly establishes the packaging mismatch, not those missing-model values.

This legacy QDS carries no verdicts

The frozen 2026-09-07 sheet predates pass_status propagation, so no status in this record reaches that sheet. Read status and interpretation from the eval; modern emitted sheets preserve source status and context.

The frozen sheet has an explicit legacy exemption

The current trust guard grandfathers this 2026-09-07 sheet before inspecting its rows. The sheet does not grade T06, and the exemption must not be read as a passed trust check or as closing anything that was open.

The harness since May

Real progress — and the check none of it performed

- ## 48-round tolerance series

Every [template] threshold that mattered here replaced by a measured band with stated preconditions.
- ## Negative-control track

0 of 22 false verdicts on the control set. Separately, 21 of 21 agent cases judged — a different denominator, since 6XVM was never judged.
- ## Gate consolidation

Negative-control headlines, governed thresholds and round-count claims are checked as data by the gate rather than trusted as prose.
- ## What none of it checked

Whether the file under evaluation is the file the report describes. No guard ties an artefact's measured model to the model its provenance names as final — which is how a categorically overstated accusation survived five adversarial review passes.

Net assessment

What the artefact supports, what it does not, what is still open

Supported

- ✓ The reported R-factors are traceable to the agent's round-7 log
- ✓ The packaged file follows the round-4 provenance trail
- ✓ The sulfate edit the report describes was correct
- ✓ The +0.0058 cross-tool offset is retained as informational

Not supported

- x A packaged copy of the reported round-7 model
- x A passing T03 gap — logged values imply 0.0501 (> 0.05)
- x The protein identity — supplied by our prompt
- x Ion assignments — both on deposited water sites
- x “T03 settled” and “refinement sound”

Open

Recover or re-run the round-7 model before any claim about the agent's final refinement · fix the upstream prompt · run and retain T02 · retain/re-run T14 residue calls · T15 needs retained content/evidence · the T16 swapped mapping is absent here (measured in the Sep21 follow-up) · geometry, waters and data quality are carried from May, not re-measured.

The agent's largest visible failure — a headline number that did not match its own shipped model — was an artefact-packaging failure. Its real failure was accepting a protein identity the benchmark handed it without checking the coordinates. This harness made the mirror-image error: it trusted a filename for five review rounds, and nearly published an overstated accusation.